

Applied Agentic AI Class · June 2026

Oldskool Mixtape Song Identification Project

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THE PROBLEM

150+ mixes. Zero tracklists.

I have 150+ recordings of live and studio electronic music mixes from 1992–1999, mixed from vinyl. Many include white labels, bootlegs, and limited-distribution remixes — with no documented tracklist.



150+

Mixes to analyze

1992

Earliest recording

1%

Tracks documented

The MC problem

Many live Drum & Bass (DnB) recordings feature an MC rapping continuously over the music — during songs, over transitions, and across the entire set. This complicates fingerprinting because:

- Vocal separation removes song vocals too, hurting fingerprint accuracy
- Stripping nothing leaves MC speech masking the audio signal
- No off-the-shelf solution handles this use case

Live set recording

Mixed signal

MC vocals

=

Music

Difficult to fingerprint

Choosing the right APIs

No Shazam API



VOCAL SEPARATION

✓ **LALAL.AI**

Orion model, stable API, batch support, 10 stem types

✗ Moises (Beta API, less mature)

SONG FINGERPRINTING

✓ **AudD**

Best DJ-mix accuracy, native MusicBrainz integration, \$5/mo Indie plan

✗ ACRCLOUD (opaque pricing, enterprise-only URL recognition)

METADATA ENRICHMENT

✓ **MusicBrainz**

Free, no key required, deep vinyl release data

✗ Discogs (auth required for useful search)

✓ **FFmpeg chunking**

AuD standard endpoint — no ngrok, works locally

✗ AudD enterprise (requires \$800/mo Pro plan)

The app asks: is there an MC?

No MC

Upload
MP3 / Video

AudD
Fingerprint original

MusicBrainz
Enrich metadata

Save
Review & export

Fastest & cheapest

MC between tracks

Upload
MP3 / Video

AudD
Fingerprint original

MusicBrainz
Enrich metadata

LALAL.AI
Optional: clean gaps

Best accuracy for song vocals

MC throughout

Upload
MP3 / Video

AudD pass 1
Fingerprint original

LALAL.AI
Strip full mix

AudD pass 2
Fingerprint instrumental

Highest coverage, uses LALAL.AI minutes

Built to be production-ready



Magic byte validation

Reads binary file header — a renamed .exe is rejected regardless of extension or MIME type



Rate limiting

5 uploads/hr · 30 polls/min · 10 AudD calls/hr per IP — prevents API credit drain



Duration cap

FFprobe measures real audio length before uploading — blocks surprise LALAL.AI billing



SSRF protection

AudD URL parameter validated as HTTPS — internal network addresses blocked



Concurrency lock

Only one LALAL.AI job at a time — second upload gets a clean 429 error



Stale file cleanup

Background job deletes temp files older than 1 hour — prevents disk buildup

A full-stack mix archiving tool

› Audio & Video input

MP3, FLAC, WAV, MP4, MOV, MKV — FFmpeg extracts audio from video automatically

› MusicBrainz enrichment

Auto-fetches label, release year, catalog number, ISRC for every identified track

› MixesDB import

Paste a MixesDB URL — scrapes full tracklist, pre-fills DJ/event/date metadata

› PDF + CSV export

Formatted PDF tracklist with metadata header, confidence indicators, ISRC data

› AudD fingerprinting

FFmpeg chunks audio into 12s clips, uploads each to AudD, deduplicates timeline

› Inline track editor

Click any track to edit — low-confidence tracks highlighted amber for review

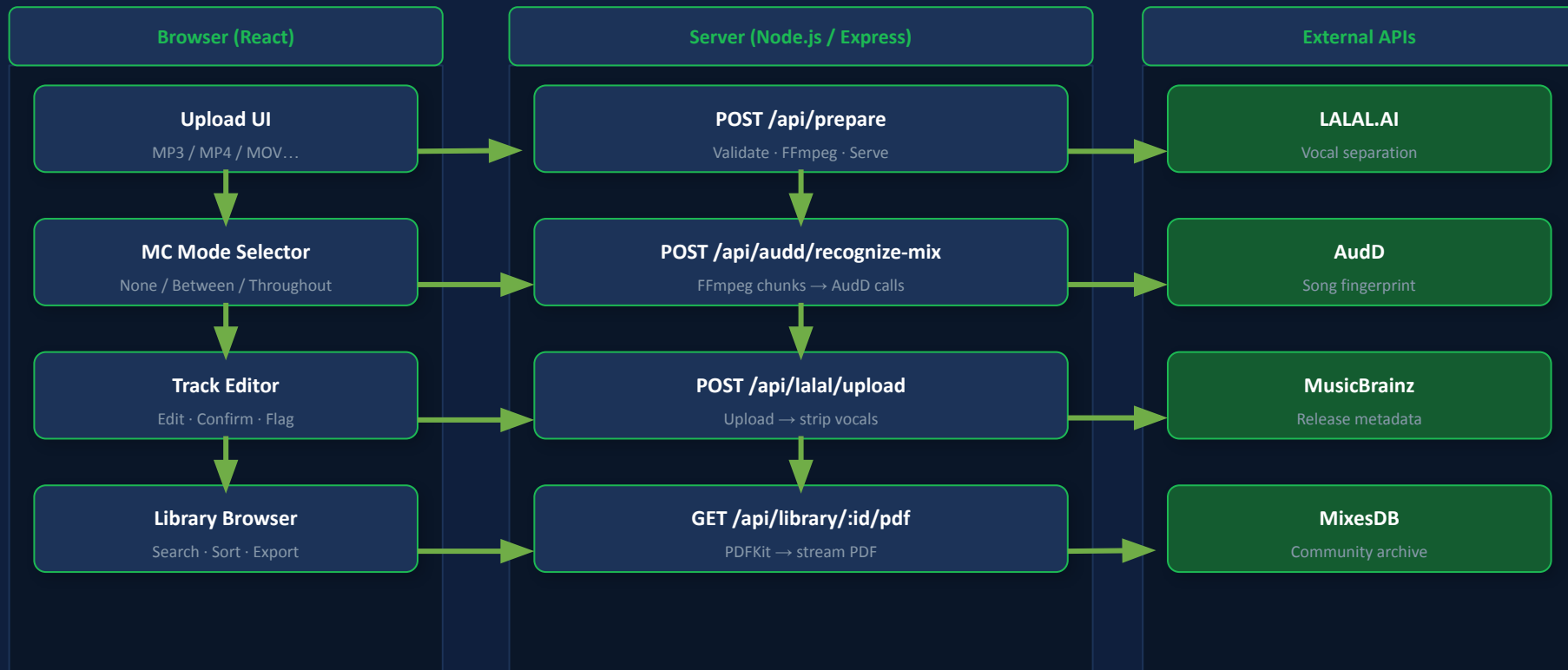
› Mix library

Persistent JSON library, searchable by DJ/event/genre, sortable by date/confidence

› Review workflow

Confirm, edit, or mark tracks unidentified — edits auto-save to library record

How the pieces fit together



USER INTERFACE & BACKEND RUNNING

Mixtape tracklist generator

Identify every song in a DJ mix. Upload audio or video — we handle the rest.



Drop a mixtape or video here

AUDIO

MP3FLACWAVM4AAAC

max 300 MB

VIDEO

MP4MOV MKVAVIWebM

max 2 GB · audio extracted automatically

Is there an MC on this mix?

Determines which processing pipeline runs.



No MC

Instrumental mix or music only. No speech to remove.



MC between tracks

MC talks over intros, outros, and transitions but not over the music itself.



MC throughout

MC talks continuously over the entire set, including during tracks.

No vocal stripping needed. AudD fingerprints the original mix directly. Fastest and cheapest pipeline.

Scan density

Every 30s — shorter = more accurate, higher API cost

10s 120s

Generate tracklist

```

[0] Mixtape server running on http://localhost:3001
[0] LALAL AI: ✓ configured
[0] AudD: ✓ configured
[0] Client secret: - disabled (set CLIENT_SECRET in .env to enable)
[0] Max audio: 300 MB
[0] Max video: 2048 MB
[0] Max duration: 90 min
[0] Pipeline: Ffmpeg chunking → api.audd.io (Indie plan)
[0]
[0] Duration: 58.9 minutes
[0] File prepared for chunking: /Users/slinky9/Documents/GitHub/Claude-Mixtape-MC-Removal-Track-ID/server/served/a18ba3967127db539e629001f481cd78.mp3
[0] AudD chunking: 118 chunks at 30s intervals over 59 min
[0] 00:00 → no match
[0] 00:30 → no match
[0] 01:00 → no match
[0] 01:30 → no match
[0] 02:00 → no match
[0] 02:30 → no match
[0] 03:00 → no match
[0] 03:30 → no match
[0] 04:00 → no match
[0] 04:30 → no match
[0] 05:00 → no match
[0] 05:30 → no match
[0] 06:00 → no match
[0] 06:30 → no match
[0] 07:00 → no match
[0] 07:30 → no match
[0] 08:00 → no match
[0] 08:30 → no match
[0] 09:00 → no match
[0] 09:30 → no match
[0] 10:00 → no match
[0] 10:30 → no match
[0] 11:00 → no match
[0] 11:30 → no match
[0] 12:00 → DJ Panini - Drumsicce (100%)
[0] 12:30 → no match
[0] 13:00 → no match
[0] 13:30 → no match
[0] 14:00 → no match
[0] 14:30 → no match
[0] 15:00 → no match
[0] 15:30 → no match
[0] 16:00 → no match
[0] 16:30 → no match
[0] 17:00 → no match
[0] 17:30 → no match
[0] 18:00 → no match
[0] 18:30 → no match
[0] 19:00 → no match
[0] 19:30 → no match
[0] 20:00 → Steve C & DJ Monita - The Razors Edge (100%)
[0] 20:30 → no match
[0] 21:00 → Prizna - Fire (Urban Shakedown Remix) (100%)
[0] 21:30 → Prizna - Fire (Urban Shakedown Remix) (100%)
[0] 22:00 → no match
[0] 22:30 → Prizna - Fire (Urban Shakedown Remix) (100%)
[0] 23:00 → Prizna - Fire (Urban Shakedown Mix) (100%)
[0] 23:30 → Prizna - Fire (Urban Shakedown Remix) (100%)
[0] 24:00 → no match
```

The underground music problem

Testing on Andy C @ Syrous — Champion of Champions (1994) revealed a fundamental database gap:

AudD identified

5

tracks from 118 chunks
(Prizna, MHC, Steve C...)

Most tracks: white labels on small UK labels never digitally distributed

MixesDB had

19

tracks — complete,
community-documented

Label, format, catalog number for every track — including dubplates
Likely identified manually by humans !

The underground music problem

Testing on Chris Sheppard - Techno 3 Still Tripping (1992) had the same issue:

Tracks found	High confidence	Needs review	Avg confidence
9	9	0	100%

Chris Sheppard - Techno 3 Still Tripping (1992)

1	19:00	Masia DJ Manolo Machaca - Masia - 北京青沫湖星文化經紀有限公司	✓	✎	100%
2	30:00	Feel So Real Dream Frequency - Get Ready For This - UMODO (Universal Music On Demand)	✓	✎	100%
3	37:30	Roughneck Project One - Best Of Techno - Volume Three - Profile	✓	✎	100%
4	43:00	Ngl CDS - Mix&Match - YRT(TOTALLY EVIL) RECORDS	✓	✎	100%
5	55:30	Ain Shams Mostafa Kelany - Ain Shams - Mostafa Kelany	✓	✎	100%
6	01:07:00	Cycles Paul Clarke Meyers - Blue Sky - OMNIMUSIC Metro Park / APM Music	✓	✎	100%
7	01:11:30	MANON B Tiakola - MANON B - M3lo World	✓	✎	100%
8	01:13:00	Grove Graumann - Domino Calling - Ir/Venir	✓	✎	100%
9	01:14:30	Tiger (Original Mix) Steve Pain - Illuminati Vol. 1	✓	✎	100%

Side A
Welcome (Dog Whistle meets the Guru mix) - B.K.S. Bekker/Kavanaugh/Sheppard ... Kavasonga/Can Win Can Music (SOCAN) (© 1992 iType)
He Never Lost His Hardcore - N.R.G. Neil Runney (Licensed from N.R.G. U.K.)
Close Your Eyes (Optikonfusion)!! - Acen Raz/Dyce ... FJR Music Ltd. (Licensed from Production House Records)
The Mayday Anthem - Westbam Westbam/Jankuhn/Dick ... Vielklang, Discoton (Licensed from Radical Records)
Raving I'm Raving (remix featuring Peter Bouncer) - Roughmix Shut Up And Dance ... Shut Up And Dance Music (Licensed from Rough Trade Records)
Closer To All Your Dreams - Rhythm Quest Rhythm Quest ... Kool Kat/Virgin Music (Licensed from Network Records)
Feel So Real (frenzy mix) - Dream Frequency Blond ... Momentum Music Ltd. (Licensed from CityBeat Records)
Dance With Power (trippier mix) - Bass Construction Pearce ... Passton Music Ltd. (Licensed from Elicit Records)
Side B
Roughneck (featuring Nanisha) - Project One Williams ... MCA Music (Licensed from Rising High Records)
Voodoo Child - Voodoo Child Richard M. Hall (BM) (Licensed from Instinct Records)
The Omen - Program II Beltram/LeFosse/Caldarone ... R&S/Program 2/Copyright Control (Licensed from R&S)
Poing - Rotterdam Termination Source Steenbergen/Scholte ... Ratal Music/Rotterdam Music (Licensed from Watta Music)
Phantom Of The Opera (maxi version) - Harajuku Andrew Lloyd Weber/Hart ... Peer Music/The Really Useful Group (Licensed from Dance Street)
Obumbratta (apocalyptic house mix) - Apotheosis Rigaux/Samoy ... Medietronic Music Group/Arcade Publ. (Licensed from Radical Records)
Let It Go - Vital Trip Ryan ... Bend 60 Music (SOCAN) (Licensed from Hi-Bias Records, Inc.)
Back To Jack Your Body (remix) - Steve "Silk" Hurley Steve "Silk" Hurley ... (Licensed from iType)

AudD identified

9 / 16

tracks from 118 chunks

But the IDs were not all correct...

2

correct identifications

What we discovered along the way

1. Test your APIs before designing the pipeline

AudD enterprise requires \$800/mo — discovered after building around it. The \$5/mo plan needs chunking, which works fine.

2. Database coverage is the real bottleneck

No amount of signal processing helps if the track isn't in the database. Community archives like MixesDB fill critical gaps.

3. Vocal stripping hurts fingerprinting

Removing MC speech also removes song vocals — the most identifying audio signal. Fingerprint the original first.

4. Agentic AI dramatically accelerates development

A full-stack app with 14 API endpoints, security guardrails, PDF generation, and a library — built in a single session.

What we can do next

→ **Discogs metadata**

After identifying a track via MixesDB, fetch catalog number, pressing info, and label from Discogs API

→ **Spotify playlist export**

One-click playlist creation from identified tracks using Spotify OAuth

→ **Auto MixesDB search**

After upload, automatically search MixesDB by filename and suggest matching sets

→ **Cloud deployment**

Replace local file serving with S3/R2 — remove ngrok dependency for production use

→ **SQLite library**

Migrate from library.json to SQLite for multi-user and higher-volume use

From vinyl to verified tracklist.

Built in one session with Claude.